Finite Element Method

FEM for Multidimensional Scalar Field Problems
MECH4103 — Spring 2026

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- The weak form for the heat conduction problem was developed in previous lectures. In matrix form, it is written as

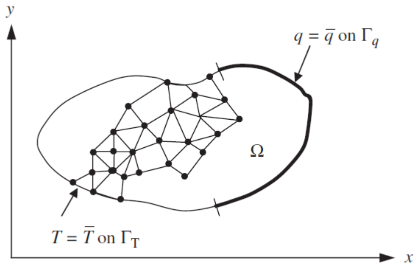
find $T(x, y) \in U$ such that:

$$\int_\Omega (\nabla w)^\mathrm{T} \mathbf{D} \nabla T \, \mathrm{d}\Omega = -\int_{\Gamma_q} w^\mathrm{T} \bar{q} \, \mathrm{d}\Gamma + \int_\Omega w^\mathrm{T} s \, \mathrm{d}\Omega \quad \forall w \in U_0, \tag{1}$$

where

$$\nabla T = \begin{bmatrix} \partial T/\partial x \\ \partial T/\partial y \end{bmatrix}, \qquad \mathbf{D} = \begin{bmatrix} k_{xx} & k_{xy} \\ k_{xy} & k_{yy} \end{bmatrix}.$$

- As a first step, the problem domain is subdivided into triangular, quadrilateral or combinations of these elements as shown in the figure. The total number of elements is denoted by n_el; the domain of each element is denoted by Ω_e.



FE mesh of the heat conduction domain with essential boundary Γ_T and natural boundary Γ_q.

- Next, the integrals in (1) are replaced by the sum of integrals over elements:

$$\sum_{e=1}^{n_{\mathrm{el}}} \left(\int_{\Omega^e} (\nabla w^e)^{\mathrm{T}} \mathbf{D}^e (\nabla T^e) \, \mathrm{d}\Omega + \int_{\Gamma_q^e} w^{e\mathrm{T}} \bar{q} \, \mathrm{d}\Gamma - \int_{\Omega^e} w^{e\mathrm{T}} s \, \mathrm{d}\Omega \right) = 0. \tag{2}$$

- The finite element approximation for the trial solution and the weight function in each element is given by:

$$\begin{aligned} T(x,y) &\approx T^e(x,y) = \mathbf{N}^e(x,y)\mathbf{d}^e = \sum_{I=1}^{n_{\mathrm{en}}} N_I^e(x,y)\, T_I^e, \quad (x,y) \in \Omega^e, \\ w^T(x,y) &\approx w^{e\mathrm{T}}(x,y) = \mathbf{N}^e(x,y)\mathbf{w}^e = \sum_{I=1}^{n_{\mathrm{en}}} N_I^e(x,y)\, w_I^e, \quad (x,y) \in \Omega^e, \end{aligned} \tag{3}$$

where n_{en} is the number of element nodes, $\mathbf{N}^e(x,y)$ is the element shape function matrix, $\mathbf{d}^e = [T_1^e \; T_2^e \; \dots \; T_{n_{\mathrm{en}}}^e]$ the element temperature matrix and $\mathbf{w}^e = [w_1^e \; w_2^e \; \dots \; w_{n_{\mathrm{en}}}^e]$ the matrix of element nodal values of the weight function.

- For an isoparametric element formulation, the shape functions are expressed in terms of natural coordinates ξ and η.
- The element nodal temperatures are related to the global temperature matrix by the **scatter matrix $\mathbf{L}^e$** (constructed exactly as in the 1D case):

$$\mathbf{d}^e = \mathbf{L}^e \mathbf{d}. \tag{4}$$

- Combining (3) and (4), we obtain relations for the trial solution and weight function in each element:

$$\begin{aligned} &\text{(a)} \quad T^e(x,y) = \mathbf{N}^e(x,y)\mathbf{L}^e\mathbf{d}, \\ &\text{(b)} \quad w^{e\mathrm{T}}(x,y) = \left(\mathbf{N}^e(x,y)\mathbf{w}^e\right)^{\mathrm{T}} = \mathbf{w}^{\mathrm{T}}(\mathbf{L}^e)^{\mathrm{T}}(\mathbf{N}^e)^{\mathrm{T}}(x,y). \end{aligned} \tag{5}$$

- The gradient field is obtained by taking the gradient of (3):

$$\nabla T^e = \begin{bmatrix} \partial T^e/\partial x \\ \partial T^e/\partial y \end{bmatrix} = \begin{bmatrix} \frac{\partial N_1^e}{\partial x} & \frac{\partial N_2^e}{\partial x} & \dots & \frac{\partial N_{n_{\mathrm{en}}}^e}{\partial x} \\ \frac{\partial N_1^e}{\partial y} & \frac{\partial N_2^e}{\partial y} & \dots & \frac{\partial N_{n_{\mathrm{en}}}^e}{\partial y} \end{bmatrix} \mathbf{d}^e.$$

- In more compact notation, the gradient is

$$\nabla T^e(x,y) = (\nabla \mathbf{N}^e(x,y))\mathbf{d}^e = \mathbf{B}^e(x,y)\mathbf{d}^e = \mathbf{B}^e(x,y)\mathbf{L}^e\mathbf{d}, \tag{6}$$

  where $\mathbf{B}^e(x,y) = \nabla \mathbf{N}^e(x,y)$.

- Applying the gradient operator to (5b), the gradient of the weight function is

$$(\nabla w^e)^{\mathrm{T}} = (\mathbf{B}^e\mathbf{w}^e)^{\mathrm{T}} = \mathbf{w}^{e\mathrm{T}}(\mathbf{B}^e)^{\mathrm{T}} = (\mathbf{L}^e\mathbf{w})^{\mathrm{T}}(\mathbf{B}^e)^{\mathrm{T}} = \mathbf{w}^{\mathrm{T}}(\mathbf{L}^e)^{\mathrm{T}}(\mathbf{B}^e)^{\mathrm{T}}. \tag{7}$$

- We will partition the global matrices as

$$\mathbf{d} = \begin{Bmatrix} \bar{\mathbf{d}}_{\mathrm{E}} \\ \mathbf{d}_{\mathrm{F}} \end{Bmatrix}, \qquad \mathbf{w} = \begin{Bmatrix} \mathbf{w}_{\mathrm{E}} \\ \mathbf{w}_{\mathrm{F}} \end{Bmatrix} = \begin{Bmatrix} \mathbf{0} \\ \mathbf{w}_{\mathrm{F}} \end{Bmatrix}.$$

The subscript 'E' denotes nodes on the essential boundaries (known values $\bar{\mathbf{d}}_{\mathrm{E}}$); subscript 'F' denotes the remaining free nodes.

- Substituting the trial solution and weight function approximations into (2) yields

$$\mathbf{w}^{\mathrm{T}} \left\{ \sum_{e=1}^{n_{\mathrm{el}}} (\mathbf{L}^e)^{\mathrm{T}} \left[\int_{\Omega^e} (\mathbf{B}^e)^{\mathrm{T}} \mathbf{D}^e \mathbf{B}^e \, \mathrm{d}\Omega \, \mathbf{L}^e \mathbf{d} + \int_{\Gamma_q^e} (\mathbf{N}^e)^{\mathrm{T}} \bar{q} \, \mathrm{d}\Gamma - \int_{\Omega^e} (\mathbf{N}^e)^{\mathrm{T}} s \, \mathrm{d}\Omega \right] \right\} = 0 \quad \forall \, \mathbf{w}_{\mathrm{F}}. \tag{8}$$

In the above, the arbitrary weight functions $w(x, y)$ have been replaced by the arbitrary parameters $\mathbf{w}_{\mathrm{F}}$, which correspond to nodes not on an essential boundary.

- We define the following element matrices.
  - **Element conductance matrix:**

$$\boxed{\mathbf{K}^e = \int_{\Omega^e} (\mathbf{B}^e)^{\mathrm{T}} \mathbf{D}^e \mathbf{B}^e \,\mathrm{d}\Omega.} \tag{9}$$

  - **Element flux matrix:**

$$\boxed{\mathbf{f}^e = \underbrace{-\int_{\Gamma_q^e} (\mathbf{N}^e)^{\mathrm{T}} \bar{q} \,\mathrm{d}\Gamma}_{\mathbf{f}_\Gamma^e} + \underbrace{\int_{\Omega^e} (\mathbf{N}^e)^{\mathrm{T}} s \,\mathrm{d}\Omega}_{\mathbf{f}_\Omega^e},} \tag{10}$$

  where $\mathbf{f}_\Gamma^e$ and $\mathbf{f}_\Omega^e$ are the element boundary and source heat flux matrices, respectively.
- The weak form can then be written as

$$\mathbf{w}^{\mathrm{T}} \left[\left(\sum_{e=1}^{n_{\mathrm{el}}} (\mathbf{L}^e)^{\mathrm{T}} \mathbf{K}^e \mathbf{L}^e \right) \mathbf{d} - \left(\sum_{e=1}^{n_{\mathrm{el}}} (\mathbf{L}^e)^{\mathrm{T}} \mathbf{f}^e \right) \right] = 0 \quad \forall\, \mathbf{w}_{\mathrm{F}}. \tag{11}$$

- The system (11) can be rewritten as

$$\mathbf{w}^{\mathrm{T}}\mathbf{r} = \mathbf{0} \qquad \forall\, \mathbf{w}_{\mathrm{F}}, \tag{12}$$

where

$$\mathbf{r} = \mathbf{K}\mathbf{d} - \mathbf{f}, \tag{13}$$

and the global matrices are assembled as:

$$\mathbf{K} = \sum_{e=1}^{n_{\mathrm{el}}} (\mathbf{L}^e)^{\mathrm{T}}\mathbf{K}^e\mathbf{L}^e, \qquad \mathbf{f} = \sum_{e=1}^{n_{\mathrm{el}}} (\mathbf{L}^e)^{\mathrm{T}}\mathbf{f}^e. \tag{14}$$

- In practice we use direct assembly rather than scatter/gather operators. We partition $\mathbf{w}$ and $\mathbf{r}$ into E- and F-nodes:

$$\mathbf{w}_{\mathrm{F}}^{\mathrm{T}}\mathbf{r}_{\mathrm{F}} + \mathbf{w}_{\mathrm{E}}^{\mathrm{T}}\mathbf{r}_{\mathrm{E}} = \mathbf{0} \qquad \forall\, \mathbf{w}_{\mathrm{F}}. \tag{15}$$

Since $\mathbf{w}_{\mathrm{E}} = \mathbf{0}$ and $\mathbf{w}_{\mathrm{F}}$ is arbitrary, the scalar product theorem gives the partitioned system:

$$\mathbf{r} = \begin{bmatrix} \mathbf{r}_{\mathrm{E}} \\ \mathbf{0} \end{bmatrix} = \begin{bmatrix} \mathbf{K}_{\mathrm{E}} & \mathbf{K}_{\mathrm{EF}} \\ \mathbf{K}_{\mathrm{EF}}^{\mathrm{T}} & \mathbf{K}_{\mathrm{F}} \end{bmatrix} \begin{bmatrix} \bar{\mathbf{d}}_{\mathrm{E}} \\ \mathbf{d}_{\mathrm{F}} \end{bmatrix} - \begin{bmatrix} \mathbf{f}_{\mathrm{E}} \\ \mathbf{f}_{\mathrm{F}} \end{bmatrix}, \tag{16a}$$

$$\begin{bmatrix} \mathbf{K}_{\mathrm{E}} & \mathbf{K}_{\mathrm{EF}} \\ \mathbf{K}_{\mathrm{EF}}^{\mathrm{T}} & \mathbf{K}_{\mathrm{F}} \end{bmatrix} \begin{bmatrix} \bar{\mathbf{d}}_{\mathrm{E}} \\ \mathbf{d}_{\mathrm{F}} \end{bmatrix} = \begin{bmatrix} \mathbf{f}_{\mathrm{E}} + \mathbf{r}_{\mathrm{E}} \\ \mathbf{f}_{\mathrm{F}} \end{bmatrix}, \tag{16b}$$

where $\mathbf{K}_{\mathrm{E}}$, $\mathbf{K}_{\mathrm{F}}$ and $\mathbf{K}_{\mathrm{EF}}$ are partitioned to be congruent with the partitions of $\mathbf{d}$ and $\mathbf{f}$.

- Consider the heat conduction problem. The domain is discretised with a **single quadrilateral element** shown in the figures.
- The boundary conditions are:
  - $T = 0$ on edges AD (left) and AB (diagonal).
  - $\bar{q} = 20$ on edge DC (top).
  - $\bar{q} = 0$ on edge CB (right).

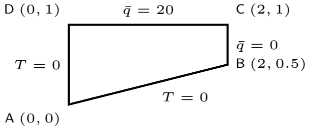


Figure 8.2 Problem definition for Example 8.1.

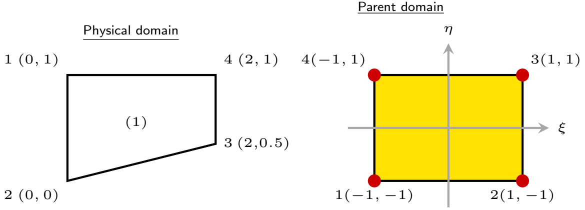


Figure 8.8 Element numbering for Example 8.2.

The element coordinate matrix is

$$\begin{bmatrix}\mathbf{x}^e & \mathbf{y}^e\end{bmatrix} = \begin{bmatrix} x_1^e & y_1^e \\ x_2^e & y_2^e \\ x_3^e & y_3^e \\ x_4^e & y_4^e \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \\ 2 & 0.5 \\ 2 & 1 \end{bmatrix}.$$

The four-node quadrilateral element shape functions in the parent domain are

$$\begin{aligned}
N_1^{4Q}(\xi,\eta) &= \frac{\xi-\xi_2}{\xi_1-\xi_2}\cdot\frac{\eta-\eta_4}{\eta_1-\eta_4} = \tfrac{1}{4}(1-\xi)(1-\eta),\\
N_2^{4Q}(\xi,\eta) &= \frac{\xi-\xi_1}{\xi_2-\xi_1}\cdot\frac{\eta-\eta_4}{\eta_1-\eta_4} = \tfrac{1}{4}(1+\xi)(1-\eta),\\
N_3^{4Q}(\xi,\eta) &= \frac{\xi-\xi_1}{\xi_2-\xi_1}\cdot\frac{\eta-\eta_1}{\eta_4-\eta_1} = \tfrac{1}{4}(1+\xi)(1+\eta),\\
N_4^{4Q}(\xi,\eta) &= \frac{\xi-\xi_2}{\xi_1-\xi_2}\cdot\frac{\eta-\eta_1}{\eta_4-\eta_1} = \tfrac{1}{4}(1-\xi)(1+\eta).
\end{aligned}$$

The gradient in the parent domain is

$$\mathbf{GN}^{4Q} = \begin{bmatrix} \partial N_1^{4Q}/\partial\xi & \partial N_2^{4Q}/\partial\xi & \partial N_3^{4Q}/\partial\xi & \partial N_4^{4Q}/\partial\xi \\ \partial N_1^{4Q}/\partial\eta & \partial N_2^{4Q}/\partial\eta & \partial N_3^{4Q}/\partial\eta & \partial N_4^{4Q}/\partial\eta \end{bmatrix} = \frac{1}{4}\begin{bmatrix} \eta-1 & 1-\eta & 1+\eta & -\eta-1 \\ \xi-1 & -\xi-1 & 1+\xi & 1-\xi \end{bmatrix}.$$

The Jacobian matrix, its determinant, and its inverse are:

$$\mathbf{J}^{(1)} = (\mathbf{GN}^{4Q})[\mathbf{x}^{(1)}\ \mathbf{y}^{(1)}] = \frac{1}{4}\begin{bmatrix} \eta-1 & 1-\eta & 1+\eta & -\eta-1 \\ \xi-1 & -\xi-1 & 1+\xi & 1-\xi \end{bmatrix}\begin{bmatrix} 0 & 1 \\ 0 & 0 \\ 2 & 0.5 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0.125\eta-0.375 \\ 1 & 0.125\xi+0.125 \end{bmatrix},$$

$$\det \mathbf{J}^{(1)} \equiv |\mathbf{J}^{(1)}| = -0.125\eta + 0.375,$$

$$(\mathbf{J}^{(1)})^{-1} = \begin{bmatrix} \dfrac{1+\xi}{3-\eta} & 1 \\ \dfrac{8}{\eta-3} & 0 \end{bmatrix}.$$

The derivatives of the shape functions with respect to the global Cartesian coordinates are

$$\mathbf{B}^{(1)} = (\mathbf{J}^{(1)})^{-1}(\mathbf{GN}^{4Q}) = (\mathbf{J}^{(1)})^{-1}\frac{1}{4}\begin{bmatrix} \eta-1 & 1-\eta & 1+\eta & -\eta-1 \\ \xi-1 & -\xi-1 & 1+\xi & 1-\xi \end{bmatrix}.$$

The conductance and flux matrices are computed using 2×2 Gauss quadrature with:

$$\xi_1 = -\frac{1}{\sqrt{3}}, \quad \xi_2 = \frac{1}{\sqrt{3}}, \quad \eta_1 = -\frac{1}{\sqrt{3}}, \quad \eta_2 = \frac{1}{\sqrt{3}}, \quad W_1 = W_2 = 1.$$

The conductance matrix is given by

$$\begin{aligned}\mathbf{K} = \mathbf{K}^{(1)} &= \int_{\Omega} (\mathbf{B}^e)^{\mathrm{T}} \mathbf{D}^e \mathbf{B}^e \,\mathrm{d}\Omega = k \int_{-1}^{1}\int_{-1}^{1} \mathbf{B}^{(1)\mathrm{T}} \mathbf{B}^{(1)} |\mathbf{J}^{(1)}| \,\mathrm{d}\xi \,\mathrm{d}\eta \\ &= k \sum_{i=1}^{2}\sum_{j=1}^{2} W_i W_j \, |\mathbf{J}^{(1)}(\xi_i, \eta_j)| \, \mathbf{B}^{(1)\mathrm{T}}(\xi_i, \eta_j) \, \mathbf{B}^{(1)}(\xi_i, \eta_j).\end{aligned}$$

Summing the contributions from the four Gauss points yields

$$\mathbf{K} = \begin{bmatrix} 4.76 & -3.51 & -2.98 & 1.73 \\ -3.51 & 4.13 & 1.73 & -2.36 \\ -2.98 & 1.73 & 6.54 & -5.29 \\ 1.73 & -2.36 & -5.29 & 5.91 \end{bmatrix}.$$

The source matrix is given by

$$\mathbf{f}_{\Omega} = \int_{\Omega^e} s(\mathbf{N}^{4\mathrm{Q}})^{\mathrm{T}} \,\mathrm{d}\Omega = \int_{-1}^{1}\int_{-1}^{1} s(\mathbf{N}^{4\mathrm{Q}})^{\mathrm{T}} |\mathbf{J}^{(1)}| \,\mathrm{d}\xi \,\mathrm{d}\eta = \int_{-1}^{1}\int_{-1}^{1} 6 \begin{bmatrix} N_1^{4\mathrm{Q}} \\ N_2^{4\mathrm{Q}} \\ N_3^{4\mathrm{Q}} \\ N_4^{4\mathrm{Q}} \end{bmatrix} (-0.125\eta + 0.375) \,\mathrm{d}\xi \,\mathrm{d}\eta = \begin{bmatrix} 2.5 \\ 2.5 \\ 2 \\ 2 \end{bmatrix}.$$

- The only contribution to the boundary flux matrix comes from the edge CD. The positive ξ direction in the parent element domain is defined from node 1 to node 2; the positive η direction points from node 1 to node 4. Therefore, edge CD in the physical domain corresponds to $\xi = -1$ in the parent domain.
- The boundary flux matrix can be integrated analytically or by one-point Gauss quadrature:

$$\mathbf{f}_\Gamma = -\int_{\Gamma^{\mathrm{CD}}} \bar{q}(\mathbf{N}^{4\mathrm{Q}})^{\mathrm{T}}\,\mathrm{d}\Gamma = -\int_{x=0}^{x=2} \bar{q}\,\mathbf{N}^{4\mathrm{Q}}(\xi=-1,\eta)^{\mathrm{T}}\,\mathrm{d}x$$

$$= -\underbrace{\frac{b-a}{2}}_{1}\bar{q}\int_{-1}^{1}\mathbf{N}^{4\mathrm{Q}}(\xi=-1,\eta)^{\mathrm{T}}\,\mathrm{d}\eta = -20\int_{-1}^{1}\begin{bmatrix}\frac{1}{2}(1-\eta)\\ 0\\ 0\\ \frac{1}{2}(1+\eta)\end{bmatrix}\mathrm{d}\eta = \begin{bmatrix}-20\\ 0\\ 0\\ -20\end{bmatrix}.$$

- The resulting right-hand-side matrix is

$$\mathbf{f}_\Omega + \mathbf{f}_\Gamma + \mathbf{r} = \begin{bmatrix} r_1 - 17.5\\ r_2 + 2.5\\ r_3 + 2\\ -18\end{bmatrix}.$$

The global system of equations is

$$\begin{bmatrix} 4.76 & -3.51 & -2.98 & 1.73 \\ -3.51 & 4.13 & 1.73 & -2.36 \\ -2.98 & 1.73 & 6.54 & -5.29 \\ 1.73 & -2.36 & -5.29 & 5.91 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \\ T_4 \end{bmatrix} = \begin{bmatrix} r_1 - 17.5 \\ r_2 + 2.5 \\ r_3 + 2 \\ -18 \end{bmatrix},$$

which yields $T_4 = -3.04$. The global temperature matrix is

$$\mathbf{d} = \mathbf{d}^{(1)} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ T_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ -3.04 \end{bmatrix}.$$

The resulting flux matrix is computed at the Gauss points:

$$\mathbf{q}^{(1)} = -k\nabla\theta = -k\mathbf{B}^{(1)}\mathbf{d}^{(1)},$$

$$\mathbf{q}^{(1)}(\xi_1, \eta_1) = -k\mathbf{B}^{(1)}(\xi_1, \eta_1)\mathbf{d}^{(1)} = \begin{bmatrix} 0.90 \\ 3.60 \end{bmatrix},$$

$$\mathbf{q}^{(1)}(\xi_2, \eta_2) = -k\mathbf{B}^{(1)}(\xi_2, \eta_2)\mathbf{d}^{(1)} = \begin{bmatrix} -2.3 \\ 19.8 \end{bmatrix},$$

$$\mathbf{q}^{(1)}(\xi_3, \eta_3) = -k\mathbf{B}^{(1)}(\xi_3, \eta_3)\mathbf{d}^{(1)} = \begin{bmatrix} 4.95 \\ 19.8 \end{bmatrix},$$

$$\mathbf{q}^{(1)}(\xi_4, \eta_4) = -k\mathbf{B}^{(1)}(\xi_4, \eta_4)\mathbf{d}^{(1)} = \begin{bmatrix} 5.81 \\ 3.60 \end{bmatrix}.$$

Why Gauss points?

The flux matrix is computed at Gauss points rather than nodes because the gradient $\nabla\theta = \mathbf{B}^e\mathbf{d}^e$ is most accurate there (superconvergence).

A critical aspect of finite element applications is *verification* and *validation*. Using the definitions of Roache:

Verification: Are the equations being solved correctly?

Validation: Are the right equations being solved?

- The first question is a question of logic and correct programming:
  - Part of the answer lies in the correctness of the elements and the weak form used in the program, the correctness of the solver and the postprocessing.
  - Part of the answer lies in the programming: are the procedures programmed correctly?
- The **patch test** has become widespread as a method for verifying finite element programs. It is extremely simple, and it is recommended even for commercial software. For a homemade code, it is essential before trying any more complicated problems.
- The patch test is based on the properties of **linear completeness** and the fact that if a finite element approximation contains the exact solution, then the finite element program must obtain that exact solution.

- In the patch test, a mesh such as shown in Figure 8.13 is made.
- The mesh can be quite arbitrary, but it is important to have **irregular elements**, as some elements are sometimes satisfactory when of regular shapes, such as rectangles, but perform quite poorly when skewed.
- **From four to eight elements are sufficient** when checking your own program with the patch test.
- The patch test is based on the properties of linear completeness and the fact that if a finite element approximation contains the exact solution, then the finite element program must obtain that exact solution.

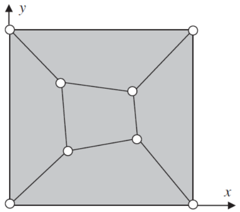


Figure 8.13 A typical finite element mesh for the patch test.

Figure 8.13 A typical finite element mesh for the patch test.

- The mesh is used to solve the heat conduction equation with prescribed temperatures:

$$T(x, y) = \alpha_0 + \alpha_1 x + \alpha_2 y, \tag{17}$$

  where α_0, α_1 and α_2 are arbitrary constants; you can set them to whatever you like, but they should all be nonzero. If you are checking your own program, it is best to give them distinctive values so that you can recognise them in the output.

- When you run the finite element program, the solution for the nodal temperatures should be given exactly by (17) with the numbers you picked for α_i, and the heat flux should be constant throughout the mesh. The values should agree to the exact values within machine precision, which can vary from 10^{-8} to 10^{-10}.

- **Why does this work?**
  - If you consider the heat conduction equation, a linear field is a solution when there are no sources. Prescribing the temperatures along the boundary by this field means that (17) satisfies the governing equation and boundary conditions.
  - As the solution to a linear problem is unique, this must be the exact solution. Furthermore, because the exact solution is included in the set of finite element approximations, the finite element solution must be the exact solution.
  - There is considerable research that shows that any element that satisfies the patch test is a **convergent element**.

- We consider a manufactured solution of the form

$$T = (r - a)^2 = x^2 + y^2 - 2a\sqrt{x^2 + y^2} + a^2,$$

defined over the domain of a square plate with a hole (Figure 8.14). For isotropic conductivity $k = 1$, the corresponding source term that satisfies the governing equation is

$$s = -\nabla^2 T = 2a \frac{1}{\sqrt{x^2 + y^2}} - 4.$$

- The **essential boundary conditions** on Γ_T are

$$T(r = a) = 0, \quad T(x = \pm b, y) = a^2 + b^2 + y^2 - 2a\sqrt{y^2 + b^2}.$$

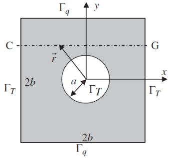


Figure 8.14 A square plate with a hole, with prescribed temperature at $x = \pm b$ and prescribed flux at $y = \pm b$.

Figure 8.14 A square plate with a hole, with prescribed temperature at $x = \pm b$ and prescribed flux at $y = \pm b$.

Example — Manufactured Solution (cont.)

- The **natural boundary conditions** on Γ_q are ($\bar{q} = -k\mathbf{n}^{\mathrm{T}}\nabla T$):

$$\bar{q}(x, y = b) = -\frac{\partial T}{\partial y}(x, y = b) = 2b\left(\frac{2a}{\sqrt{x^2 + y^2}} - 1\right),$$

$$\bar{q}(x, y = -b) = -\frac{\partial T}{\partial y}(x, y = -b) = 2b\left(1 - \frac{2a}{\sqrt{x^2 + y^2}}\right).$$

- Figure 8.15 depicts the coarsest and the finest meshes considered in the convergence studies and the temperature distribution in both meshes.

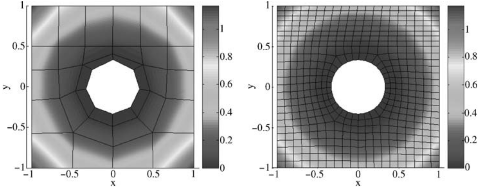


Figure 8.15 Temperature distribution for the coarsest (34-element) and the finest (502-elements) meshes.

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- Figure 8.16 compares the temperature distribution along the line GG′ obtained with the two meshes against the exact solution.
- Figure 8.17 depicts the log–log plot of the error in the L_2 and energy norms, with a linear approximation obtained by linear least squares regression.
- It can be seen that the slopes approximately equal **1** and **2** in the L_2 and energy norms, respectively, closely matching the theoretical values. Identical results were found in one dimension.

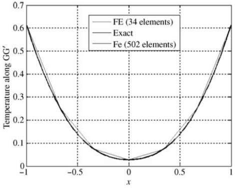

Figure 8.16 Temperature along the line GG′ for the coarsest (34-element) the finest (502-elements) meshes.

Figure 8.16 Temperature along line GG′ for the coarsest (34-element) and finest (502-element) meshes.

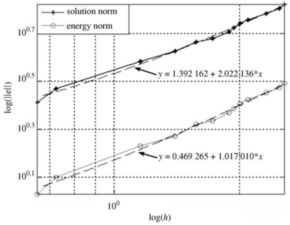

Figure 8.17 Convergence in L_2 and energy norms.

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Next: Mesh Generation

Thanks for your attention!